

SGraph Send

Zero-Knowledge File Sharing



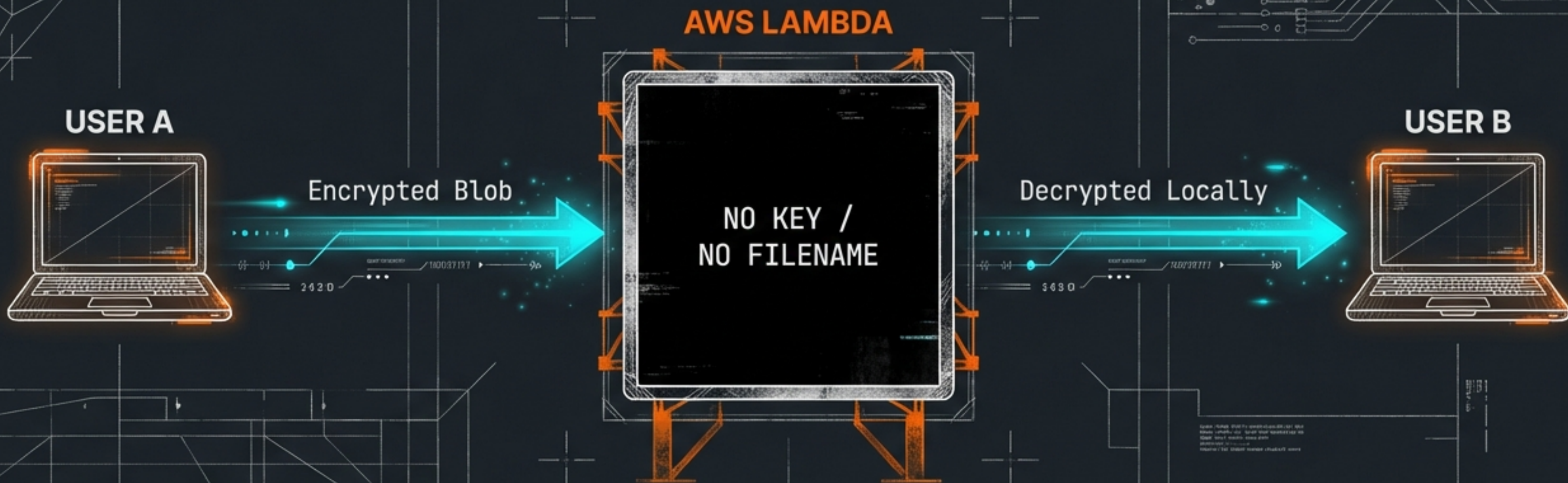
BUILT BY AI AGENTS

DEPLOYED TO AWS LAMBDA

MVP v0.2.15 | 12 FEB 2026

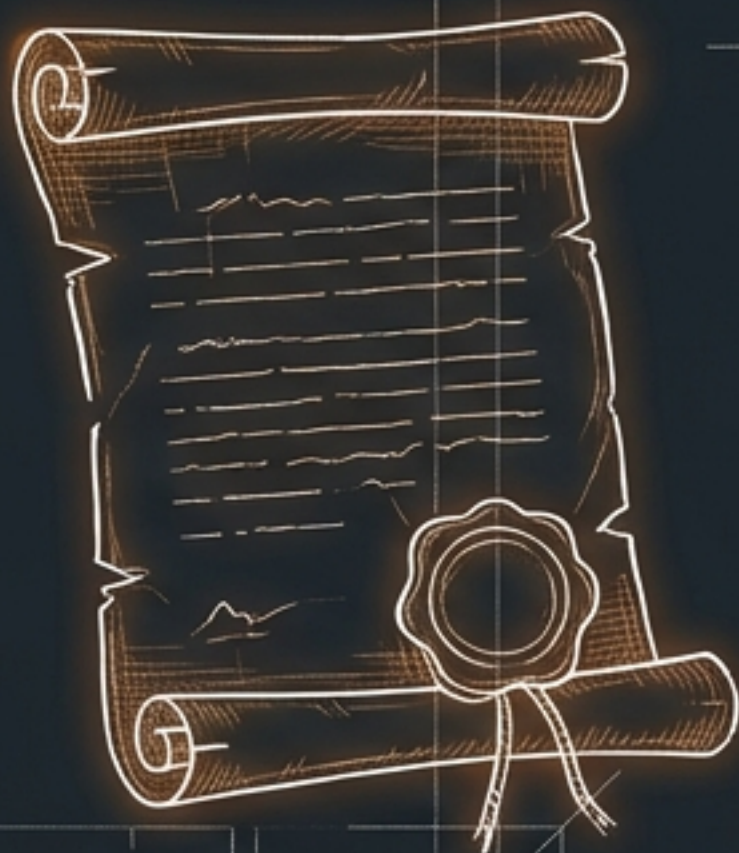
12 Feb 2026: The First Verified Transfer

Architecture verified: Server received opaque blob. Decryption occurred only on client devices.



Privacy by Architecture, Not Policy

STANDARD MODEL



Privacy by Policy

Trust based. "We promise not to look."

The server, at no point, possesses the decryption key,
the file name, or the plaintext content.

SGRAPH MODEL



Privacy by Design

Math based. "We are architecturally incapable of looking."

Zero-Knowledge in Practice



1. Browser generates
AES-256-GCM key.
Key remains on device.



2. Client-side
Encryption.
File becomes opaque
binary blob.



3. Transit.
Only blob + metadata
travel to server.

Encrypted Blob

Metadata



4. Receiver decrypts
locally using
shared link key.

NOTE: Key space (2^{256}) >
atoms in the universe.

Transparency as a Product Feature

Instead of a hidden policy,
we show a live data receipt.

Source: `send-transparency.js`

SERVER RECEIPT:

[CAPTURED]

- > IP Hash: `sha256(...)`
- > Timestamp: `2026-02-12T14:00:00Z`
- > File Size: `14MB`

[NOT CAPTURED]

- > File Name: `NULL`
- > Content: `ENCRYPTED`
- > Key: `NULL`

Cryptography Specifications

ALGORITHM	AES-256-GCM (Galois/Counter Mode)
INITIALISATION VECTOR	96-bit (12 bytes), random per file
KEY FORMAT	Base64url-encoded, 44 characters
API STANDARD	Web Crypto API (Native)

SECURITY NOTE:

Breach Consequence: NULL. Without the key, server data is computationally useless.

Serverless Infrastructure

Python 3.12
(FastAPI via Mangum)

AWS Lambda
(Function URLs / No API Gateway)

Memory-FS Abstraction
(Storage)

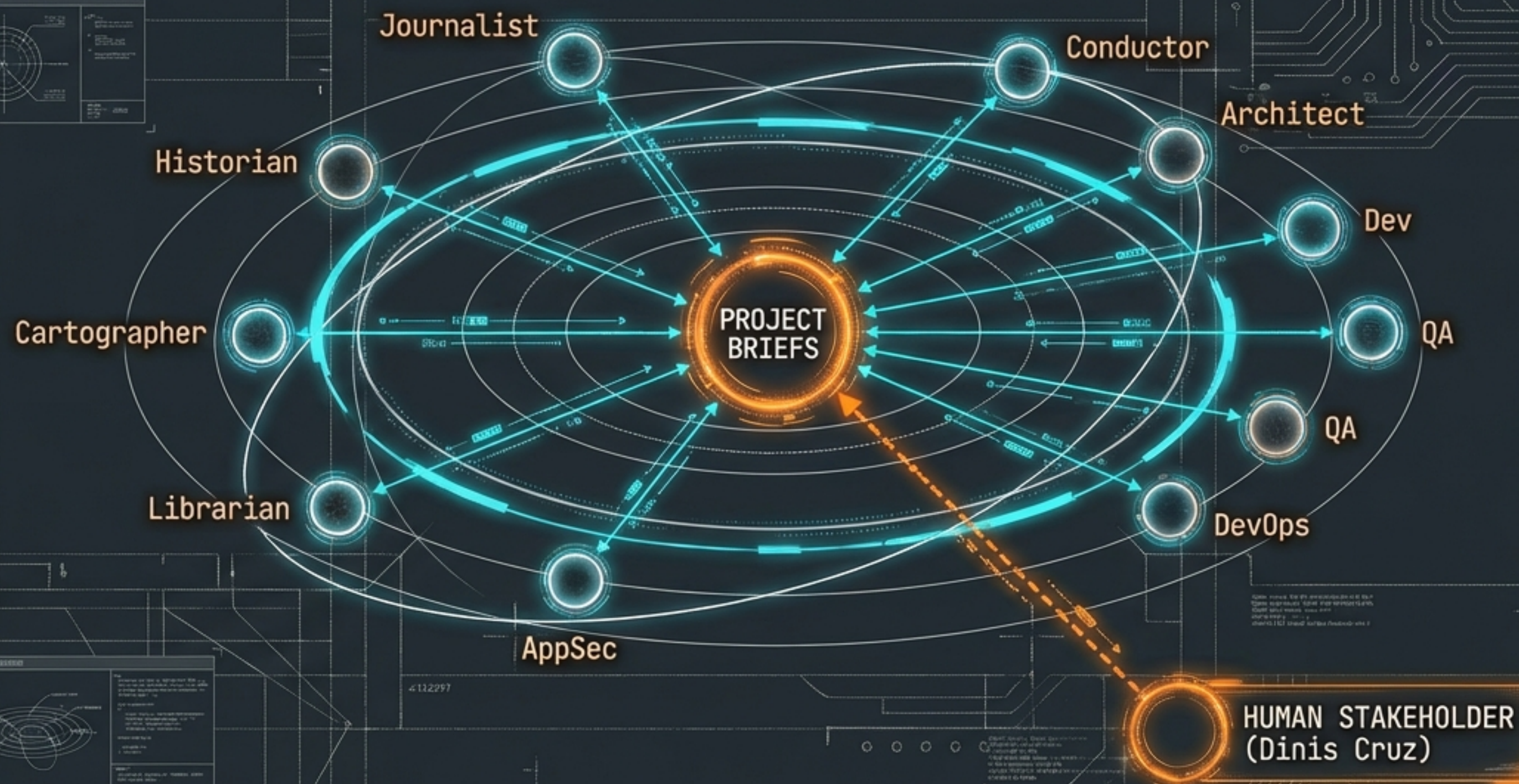
~2,000

Lines of Code
(Total Repo)

~1.5
Minute

Deployment Time

Built by a Coordinated System of Agents



Structured Autonomy & Verification



VERIFICATION RIGOUR

- ✓ 44 Unit Tests (In-Memory)
- ✓ Zero Mocks Used
- ✓ AppSec: Encryption Model Verified
- ✓ QA: Schema & Route Validation

From Brief to Live in 12 Days

Early Feb -
Project Brief

10 Feb -
Infrastructure
Brief

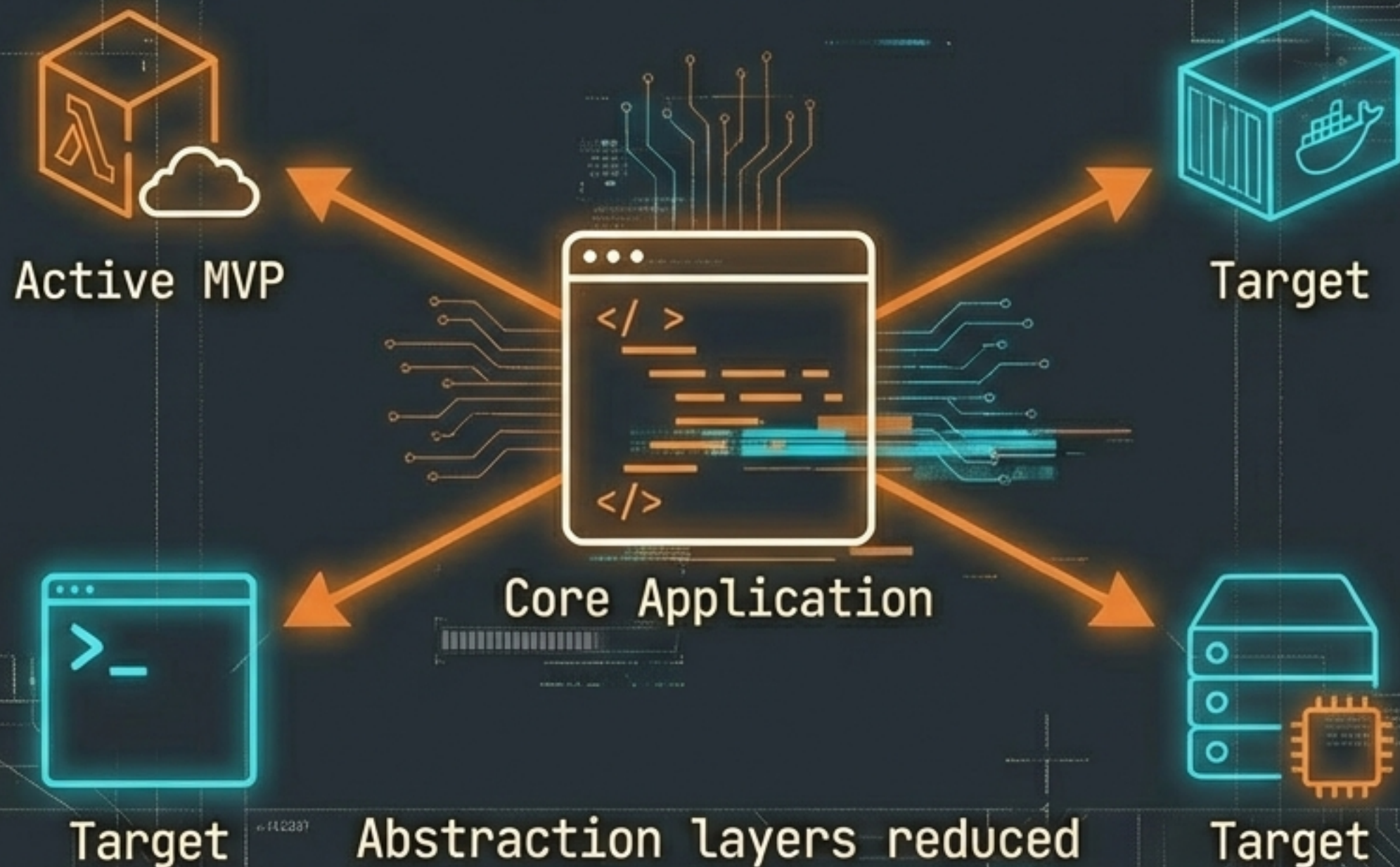
11 Feb - Backend
& Frontend Build

12 Feb - LIVE MVP
(CI/CD Automated)

Total Velocity: 12 Days

Deployment Strategy

Write Once, Deploy Everywhere



Abstraction layers reduced
infrastructure work by 40%.

Immediate Roadmap

PERSISTENCE

Swap In-Memory for S3 Backend via Memory-FS abstraction.



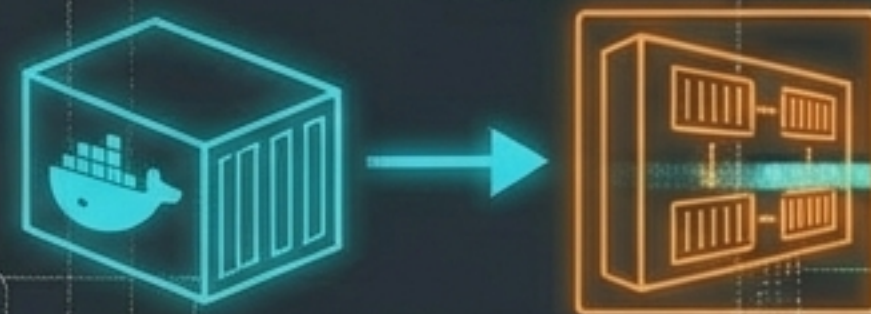
HARDENING

Implement Type_Safe classes from osbot-utils.



EXPANSION

Deploy to Docker and ECS Fargate.



Summary of Verified Claims

- [VERIFIED] End-to-end encryption works in browser
- [VERIFIED] Server remains blind to content and keys
- [VERIFIED] Transparency mechanism displays live data
- [VERIFIED] CI/CD pipeline deploys in ~1.5 mins
- [VERIFIED] Zero-Knowledge is architecturally enforced



File Sharing. Simple. Private by Default.

Zero-Knowledge validated on
Serverless Infrastructure.

> Read the Code (GitHub: [sgraph-ai-app-send](#))

> Read the Story (LinkedIn)